IAGO VALE

Data Scientist & AI Engineer

BASED IN Vila Velha, ES · Brazil (UTC–3) CITIZENSHIP U.S. · no visa sponsorship needed
imaurevale@gmail.com · linkedin.com/in/iago-vale-53299a190

SUMMARY

I'm a Data Scientist and AI Engineer. I build AI agents that automate real business operations, from data pipelines to construction estimating and client communication, and I measure whether they actually work. My foundation is statistics: a B.Sc. from UFES and 5+ years in data as a Data Scientist and Data Analyst, from supervised and unsupervised models to BI.

EXPERIENCE

AI Engineer, Grupo Carone Sep 2026 – Present

- Building an agentic ETL platform where orchestrated AI agents (LangGraph + n8n) generate and maintain SQL/Python data pipelines for sales, finance, inventory/procurement and operations.
- Grounding agents with RAG over the data dictionary, documentation and business rules, plus semantic and procedural memory, so every transformation follows company definitions and reuses patterns that already worked.
- Gating every generated pipeline with evals before it ships, and tracing every agent run end to end (observability).
- Fine-tuning a model to generate SQL/Python that follows the company's schemas and coding standards.
- Designing orchestration and monitoring in n8n: agents detect failed loads, reprocess them and alert the data team.
- Applying agent architecture and harness engineering (tools, context and execution loop) to make agent-generated pipelines reliable enough for production.

AI Automation Engineer – Estimating & Proposals, Akaza Builders · USA (remote) Nov 2025 – Aug 2026

- Designed and deployed AI agents that automate construction estimating and proposal drafting.
- Built a RAG knowledge base over past bids, pricing sheets, supplier and subcontractor quotes, project specs and drawings, so agents draft estimates grounded in the company's own data.
- Modeled the construction domain as an ontology (trades, materials, units and cost items) so agents map every spec item to a priceable item.
- Applied prompt engineering, agent harness design, evals and observability to keep outputs consistent enough for daily production use.
- Saved 15+ hours of estimating and admin work per week.

AI Automation Engineer – Client Operations, Realplan Construction Inc · USA (remote) Aug 2024 – Oct 2025

- Built AI agents that automate client and subcontractor communication: email triage, drafted replies, follow-ups and scheduling.
- Implemented RAG over email history, project documents and company policies so agents answer questions with accurate, project-specific context.
- Modeled the business as an ontology (jobs, clients, subcontractors and request types) so triage links every email to the right job, with evals and per-run tracing.
- Saved 10+ hours of administrative work per week.

Data Analyst, Grupo Carone Oct 2023 – Sep 2024

- Created data visualizations and dashboards in Qlik Sense.
- Built ETL and data wrangling in SQL, and analyses that generate actionable insights for decision-making.
- Automated reporting processes to improve efficiency.
- Mentored other data analysts.

TOOLS: SQL, ETL pipelines, Qlik Sense

Risk Analyst, Autoglass Apr 2023 – Sep 2023

- Priced insurance assets by evaluating and categorizing vehicles by brand, model and year.
- Studied portfolio behavior: customer profiles, segmentation, consumption patterns and the impact of pricing strategies.
- Built business-opportunity reports and sales proposals in Excel and Tableau.

TOOLS: Excel, Tableau

Data Scientist, Apex Partners Aug 2021 – Apr 2023

- Created ETL processes and performed data wrangling in Python and SQL.
- Ran supervised and unsupervised learning analyses, including clustering, in Python and R.
- Produced data visualization dashboards in Power BI.

TOOLS: Python, R, SQL, Supervised learning, Unsupervised learning, Power BI

SELECTED PROJECTS

Agentic ETL platform, Grupo Carone · *In development* 2026

A platform where orchestrated AI agents generate, test and maintain the SQL/Python pipelines behind sales, finance, inventory/procurement and operations.

STACK: LangGraph, n8n, LLMs, RAG, Semantic & procedural memory, Fine-tuning, Evals, Observability, Python, SQL

Estimating & proposal agents, Akaza Builders · USA · *Shipped* 2025–2026

Agents that turn specs and drawings into priced estimates and drafted proposals for a U.S. builder. *Estimating & admin work saved: 15+ h/week.*

STACK: AI Agents, Ontology, RAG, Semantic & procedural memory, Fine-tuning, Evals, Observability, n8n, LLMs, Prompt engineering

Client communication agents, Realplan Construction · USA · *Shipped* 2024–2025

Agents that handle client and subcontractor communication for a U.S. builder: triage, replies, follow-ups and scheduling, with the context of each job. *Admin work saved: 10+ h/week.*

STACK: AI Agents, Ontology, RAG, Semantic & procedural memory, Evals, Observability, n8n, LLMs, Email automation

SKILLS

AI Engineering: AI agents, Agent architecture, RAG, Prompt engineering, Harness engineering, Evals, Observability, Ontology, Semantic & procedural memory, Fine-tuning.

LLMs & orchestration: n8n, LLMs, Multi-agent orchestration, LangGraph; *familiar: LangChain, LangSmith, Langflow*.

Data science: Python, R, Statistical modeling, Supervised learning, Unsupervised learning, Time series.

Data engineering: SQL, ETL pipelines, Data wrangling.

BI & visualization: Qlik Sense, Power BI, Excel, Tableau, R Shiny.

EDUCATION

B.Sc. in Statistics, UFES — Universidade Federal do Espírito Santo 2018–2023

RESEARCH & VOLUNTEERING

Undergraduate research scholarship, Statistics, UFES · mixed models with time series on air pollution and children's respiratory disease in Greater Vitória, in R and R Shiny 2019–2020

Vice-President, Statistics Student Association, CAEST · UFES 2019–2020

IT Advisor, Educação Para a Liberdade (NGO) · set up the Bitrix24 CRM and trained the team 2020–2021

LANGUAGES

English & Portuguese (native) · Spanish (fluent)